GRADIENT DESCENT + LINEAR REGRESSION

M observations, N features; $x_{io} = 1$. **Model:** $z_i = w^T x_i = w_o + \sum_{j=1}^{N} w_j x_{ij}$. Training minimizes J over w; x, y are fixed.
$J = MSE = (1/M) \sum_i (y_i - z_i)^2$
$\partial J/\partial w_j = -(2/M) \sum_i (y_i - z_i) x_{ij}$
$\nabla J = (2/M) X^T(Xw - y)$
Why −2 and x_{ij}? Chain rule: derivative of the square gives $2(y_i - z_i)$; the inside derivative is $-x_{ij}$. For w_o, $x_{io} = 1$.
GD: $w_{k+1} = w_k - \eta \nabla J(w_k)$
Initialize w → gradient → update **all coordinates using old w** → repeat until $\|w_{k+1} - w_k\|_2 <$ tol (or iteration limit).
$-\nabla J$ = steepest local decrease; ∇J points uphill. Its norm is **not** distance to the optimum; it need not point straight at it.
η too big: overshoot/diverge; **too small:** slow. Adaptive example: $\eta_k = 1/k$. **tol↓:** more accurate/more steps; **tol↑:** faster/less accurate. Course: η, tol are **algorithm settings, not model hyperparameters**; no CV.
Scale: unequal feature magnitudes → elongated bowl → zig-zag/slow GD. Scaling improves conditioning, not convexity.
Convex: $\nabla^2 J = (2/M) X^T X \geq 0$. Full column rank → strictly convex bowl, unique global minimum.
Least squares: $\hat{w} = (X^T X)^{-1} X^T y$
Closed form if invertible; costly at scale. GD is numerical.

REGULARIZATION: FIT + PENALTY

Ridge: $J = MSE + \lambda \|w\|_2^2$
Lasso: $J = MSE + \lambda \|w\|_1$
ElasticNet: $J = MSE + \lambda_1 \|w\|_1 + \lambda_2 \|w\|_2^2$
$\|w\|_1 = \sum_j |w_j|$; $\|w\|_2^2 = \sum_j w_j^2$. **λ↑:** more shrinkage/regularization, less overfit; too large → underfit. λ is a **model hyperparameter**.
Ridge: smooth; closed form or plain GD. $\nabla J = \nabla MSE + 2\lambda w$.
Lasso: sets some weights **exactly to 0** (feature selection). |w| is nondifferentiable at 0 → no plain GD/general closed form; use subgradient/coordinate descent. ElasticNet has the same issue.
ElasticNet: $\lambda_1 = 0$ gives Ridge; $\lambda_2 = 0$ gives Lasso. Ideal tuning including endpoints can match either's best selection score; tuning two parameters costs more. Neither Ridge nor Lasso always wins.
Slides penalize w_o. If the intercept is excluded, omit its penalty and penalty-gradient term.

LOGISTIC REGRESSION + DERIVATIONS

$p_i = P(y_i = 1 \mid x_i) = \sigma(z_i) = 1/(1 + e^{-z_i})$
$\sigma'(z) = e^{-z}/(1 + e^{-z})^2 = \sigma(z)[1 - \sigma(z)]$
Split: $[1/(1 + e^{-z})] \cdot [e^{-z}/(1 + e^{-z})]$. Factor 1 = σ(z); factor 2 = $[(1 + e^{-z}) - 1]/(1 + e^{-z}) = 1 - \sigma(z)$.
$J = BCE = -(1/M) \sum_i [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)]$
y = 1: loss $-\ln p$. **y = 0:** loss $-\ln(1 - p)$. Confident correct → 0; confident wrong → ∞. Independent Bernoulli likelihood: $L = \prod_i p_i^{y_i}(1 - p_i)^{1-y_i}$; BCE = −ln L/M.
$\partial \ell_i/\partial p_i = (p_i - y_i)/[p_i(1 - p_i)]$
$\partial p_i/\partial w_j = p_i(1 - p_i) x_{ij}$
$\Rightarrow \partial J/\partial w_j = (1/M) \sum_i (p_i - y_i) x_{ij}$
$\Rightarrow \nabla J = X^T(p - y)/M$
Why BCE? Convex in w; sigmoid-MSE is nonconvex. No closed form → GD in class. Regularized classification = BCE + the same penalties.
Predict 1 if $p \geq$ threshold ($0.5 \Leftrightarrow z \geq 0$). Multi-class: categorical cross-entropy. Perfect separation: unregularized weights can diverge; **no finite max-margin convergence**.

HIERARCHY + LARGE DATA

Agglomerative: singletons → merge. **Divisive:** one cluster → split. Dendrogram height = distance; a horizontal cut sets clusters. Good for small data; costly/messy at scale. Mini-batch K-Means/BIRCH help large data.

SUPPORT VECTOR MACHINES

Large margin aids generalization. $f = b + v^T x$; v = slopes (**norm excludes b**). Labels: **−1, +1**.
Boundary: f = 0; parallel support planes: $f = \pm a$
Point-to-boundary distance = $|f(x)|/\|v\|_2$
Full margin = $2a/\|v\|_2$; normalize a = 1
Derive: subtract support-plane equations → projected separation × $\|v\|$ = 2a. Max margin ⇔ min $\|v\|$ **subject to no margin violations**. Without constraints: v = 0.
Hard margin: min $\|v\|$; $y_i f(x_i) \geq 1$ for all i
Requires separability. Support vectors touch the margin; soft-margin violations can also be support vectors.
Soft margin: $H = \sum_i \max(0, 1 - y_i z_i) + \lambda \|w\|_2^2$
Score $z_i = w^T x_i$; signed margin $m_i = y_i z_i$. **Hinge:** $h_i = \max(0, 1 - m_i)$.
m ≥ 1: correct, outside/on margin → loss 0.
0 < m < 1: correct but inside margin → 1 − m.
m = 0: on boundary → 1.
m < 0: misclassified → loss > 1.
λ↑: wider margin, more regularization/tolerance of violations; too high → underfit. **λ↓:** narrower margin, less regularization; too low → overfit. **$C \propto 1/\lambda$** acts oppositely.
Convex, nonsmooth at m = 1 → GD extensions (or quadratic programming). Kernel SVM handles nonlinear boundaries; costly for big data. Scale features.
SVC needs extensions for probabilities. Soft margins tolerate outliers; logistic optimizes BCE.
SVR: regression; course: "outermost" support vectors. Zero loss inside ε-tube; penalize observations outside.

MULTI-CLASS: ONE TARGET, C CLASSES

KNN/trees: multiple classes by votes; odd K can still tie (2–2–1). Binary logistic/SVM need extensions.
OvR = OvA: C models, c vs rest; pick largest positive-class score. Several say 1 → most confident positive; all 0 → least confident negative.
OvO: train **$C(C - 1)/2$** models, one per pair using only that pair's data. Majority vote; confidence breaks ties. C = 6 → 15 models.
OvR: linear count, more imbalance (under/oversampling can help). **OvO:** quadratic count, smaller datasets; can be faster for few classes/large data. Neither always wins.

METRICS: READ THE MATRIX AXES FIRST

With **rows = actual, columns = predicted**, for class c:
TP: diagonal cell; **FP:** other entries in its column;
FN: other entries in its row; TN = M − TP − FP − FN.
$P_c = TP/(TP + FP)$ = diagonal / column sum
$R_c = TP/(TP + FN)$ = diagonal / row sum
Macro $P = (1/C)\sum_c P_c$; macro $R = (1/C)\sum_c R_c$
Macro = equal class weights. Bad minority performance hurts; not a support-weighted average.
Micro $P = \sum TP/(\sum TP + \sum FP)$
Micro $R = \sum TP/(\sum TP + \sum FN)$
Micro = pool counts first. Majority dominates, can hide minority failure. Single-label multi-class, all classes: **micro P = micro R = accuracy**.
Accuracy: sum diagonal / M; uniform-guess baseline 1/C. **ROC-AUC:** one OvR curve/AUC per class.

MULTI-LABEL: L TARGETS PER OBSERVATION

Binary relevance: L independent models; ignores target correlations. **Label powerset:** one class per label combination; classification only, up to 2^L classes → impractical.
Classifier/regressor chain: L models, adding previous target predictions to x; uses target correlations. Order matters; errors can propagate.
Hamming loss = $(1/(ML)) \sum_i \sum_l I(y_{il} \neq \hat{y}_{il})$
Wrong labels / all entries. Lower is better; 0 = perfect. For continuous responses, pool entries for RMSE/MAE.
1 − Hamming loss = labelwise accuracy (slides call it "Jaccard"). Standard Jaccard = TP/(TP + FP + FN), comparing positive-label sets.

K-MEANS + LLOYD'S ALGORITHM

No target y. Input **K**; hard labels, no probabilities/noise. Course: Gaussian, diagonal Σ (uncorrelated). Favors round clusters. Scale.
Assignment: $c_i = \arg\min_c \|x_i - \mu_c\|_2$
Update: $\mu_c = (1/n_c) \sum_{i:c_i=c} x_i$
Initialize labels → update first; centroids → assign first. Alternate until stable. Different starts → different **local optima**; restarts/spread-out seeds (K-Means++) help.
Inertia: $I = \sum_i \|x_i - \mu_{c_i}\|_2^2$
$I \geq 0$; optimum nonincreasing as K↑; K = M → 0. **Elbow:** K where decrease slows sharply. Lowest inertia alone overselects K; elbow can be unclear.

K-MEANS EXTENSIONS

K-Modes: categorical features; centroid = featurewise **mode**, assignment by **matching dissimilarity** $d_{cat}(x, u) = \sum_j I(x_j \neq u_j)$, the number of mismatches.
K-Prototypes: mixed numeric + categorical; mean for numeric coordinates, mode for categorical.
$d_{total} = d_{Euclidean,numeric} + \gamma d_{cat}$ [course formula]
γ balances continuous distances vs integer mismatch counts so neither dominates. E.g. [red, large, metal] vs [red, small, wood] → 0 + 1 + 1 = **2** mismatches.
K-Medoids: actual observation minimizing total distance to its cluster's points. Assign to nearest medoid. More robust to outliers than means; more computation.

GMM: SOFT CLUSTERS + EM

Input C; μ_c: N-vector mean, Σ_c: N×N covariance, π_c: weight. Full Σ allows tilted/correlated clusters. Still Gaussian; no noise label. Scale features (course).
$p(x) = \sum_c \pi_c \varphi_c(x)$; $\pi_c \geq 0$, $\sum_c \pi_c = 1$
$\varphi_c(x)$ = Gaussian density $N(x \mid \mu_c, \Sigma_c)$.
E-step: parameters → probabilities (Bayes). C responsibilities per point, summing to 1.
$\tau_{ic} = P(c \mid x_i) = \pi_c \varphi_c(x_i) / \sum_r \pi_r \varphi_r(x_i)$
M-step: probabilities → parameters (maximize expected complete-data log-likelihood). Soft count $n_c = \sum_i \tau_{ic}$.
$\mu_c = \sum_i \tau_{ic} x_i / n_c$; $\pi_c = n_c/M$
$\Sigma_c = \sum_i \tau_{ic}(x_i - \mu_c)(x_i - \mu_c)^T / n_c$
Use **new μ** in Σ; denominator n_c, not $n_c - 1$. Larger τ → more influence. **π_c = average membership over points**; τ_{ic} = one point's membership.
Initialize → E → M → E … until stable. E gets parameters from **previous M**; M uses latest E probabilities. Hard label = largest τ.
E ↔ Assignment; M ↔ Update. Practice: hard 0/1 probabilities + diagonal Σ ⇒ Lloyd is **true in course**. Exact Euclidean form: shared $\sigma^2 I$, equal weights (or small-variance limit).
BIC = −2 ln L + q ln M; $L = \prod_i p(x_i)$
Lower BIC wins. More clusters: likelihood↑ (term 1↓), parameter count q↑ (term 2↑). Balances fit vs complexity.
Generative: estimates source density; generate by choosing c via π, then sampling x from its Gaussian.

DB-SCAN + SILHOUETTE

DB-SCAN: nonparametric, arbitrary shapes, outliers; **no K input**. Radius ε (= tol); min_samples sets density (course: min cluster size). Grow via nearby points; isolated/sparse groups → noise. Scale.
tol too large: joins distinct clusters. **Too small:** fragments clusters/more noise. min_samples↑ rejects more sparse groups. Sensitive to radius; varying densities are hard → OPTICS/HDBSCAN (more computation).
Silhouette: $s = (b - a)/\max(a, b)$, in [−1, 1]
Slides: a = avg point-to-own-centroid distance; b = avg centroid-pair distance. 1 = separated; 0 = overlap; negative = poor assignment. Applicable across clustering models; inertia/elbow poor for DB-SCAN.
Standard: a = mean distance to own-cluster peers; b = nearest other cluster's mean distance; average s_i. Follow the question's convention.

Q1 QUESTIONS + CORRECT ANSWERS

1(a) In each step of gradient descent, why is the step taken in the opposite direction of the gradient vector (i.e. $-\nabla$MSE)?

Answer: The gradient vector points in the direction of steepest increase, so moving opposite to it decreases the cost fastest.

1(b) Why is the MSE cost not used to train a Logistic Regression model?

Answer: With the sigmoid inside it, the MSE cost is no longer a convex function, so gradient descent becomes very inefficient.

1(c) If a classification dataset has six classes and a binary classifier is trained on it with the One-vs-One extension, how many binary classifiers are trained?

Answer: 15

1(d) A regression dataset has two numeric features: one ranges from 0 to 1, the other from 0 to 100,000. Without scaling, what happens when gradient descent is run on a Linear Regression model trained on this data?

Answer: The cost function turns into an "elongated bowl", so the algorithm zig-zags and becomes sensitive to the initialization point and the learning rate.

1(e) For a Support Vector Machine (SVM) classifier, which of the following optimization problems does maximizing the margin lead to?

Answer: Minimize $\|w\|$, the norm of the parameter vector, subject to no observations lying inside the margin.

1(f) For a Gaussian Mixture Model (GMM) and the Expectation-Maximization (EM) algorithm that trains it, what do the cluster weights (π_c's) represent for clusters c = 1, …, C?

Answer: The average probability, across all observations in the data, of belonging to cluster c.

REVIEW: MODEL SETUP + GRADIENT DESCENT

Parametric model: equation + cost + training algorithm. **GD:** approximate ("close enough"), applies to differentiable costs, efficient for big data. **Least squares:** exact/algebraic for linear and Ridge; costly at scale.

GD step length = $\eta\|\nabla J\|$.

Gradient shrinks near the minimum → smaller steps. Choose η, tol by rules of thumb/trial and error.

Scaling: problems worsen with more features/larger magnitude gaps; initialization and η affect convergence.

Penalty: balances fit against large weights.

REVIEW: SVM DETAILS

Linear SVM is **parametric, interpretable**; classification/regression, nonlinear with kernels. Support vectors: classifier = **innermost/closest to boundary**; regressor = outermost (course).

Extreme λ: very low → vanishing margin/overfit; very high → weights toward 0, enormous (course: infinite) margin/underfit.

SVC needs extensions for **probability thresholds**.

OvR: class c → 1, rest → 0; confidence = logistic probability or SVM signed boundary distance.

Large feature values dominate Euclidean distances. K-Means/GMM assign all points to K clusters; DBSCAN automatically labels noise.

Elbow axes: x = K, y = inertia; K = 1 gives maximum inertia, K = M gives 0. Medoid: minimize sum/average distance (equivalent).

EM stop: μ, Σ, π, τ stable. GMM fits a PDF/generates data; other models in the review focus on values/classes/clusters (course: "discriminative").

Q2 SELECTED ANSWERS

2(a) Which of the following statements are **TRUE** about Lasso Regression?

Correct statements:
- It cannot use plain gradient descent, so extensions such as subgradient or coordinate descent are needed.
- It can drive some trained parameter values to exactly zero, and performs feature selection that way.

2(b) Which of the following statements are **FALSE** about precision and recall for a multi-class classification task?

Select these FALSE statements:
- Micro precision/recall treats every class as equally important.
- Micro precision/recall is defined to reflect a sharp decrease in performance due to a poorly predicted minority class.
- Macro precision/recall takes a weighted average of all (One-vs-Rest/binary) precision/recall values calculated for each class present in the data.

2(c) Which of the following statements are **TRUE** about shortcomings of a K-Means clustering model that are addressed by other clustering models?

Correct statements:
- DB-SCAN addresses finding possible outliers in the data.
- DB-SCAN addresses non-Gaussian underlying sources.

2(d) Which of the following statements are **FALSE** about the E-Step of the EM algorithm?

Select these FALSE statements:
- The current E-Step receives the current model parameters from the previous E-Step.
- Given the necessary assumptions, the E-Step of the EM algorithm becomes equivalent to the Update step of Lloyd's algorithm.

Q3 TRUE / FALSE - OFFICIAL ANSWERS

3(a) TRUE. For gradient descent, a smaller tolerance gives parameters closer to the closed-form (Least Squares) solution but takes more steps to converge.

3(b) FALSE. If two classes in a binary classification task are perfectly separable, a Logistic Regression model trained with gradient descent converges to the decision boundary with the largest margin between both classes.

3(c) FALSE. The classifier chain approach is applicable to a multi-class classification task with only one response/target variable.

3(d) FALSE. In the Hinge cost of an SVM model, increasing λ, the hyperparameter, shrinks the margin and reduces regularization.

3(e) TRUE. For a K-Means clustering model, inertia decreases monotonically as K (number of clusters) increases, and it equals 0 when K equals the number of observations.

3(f) TRUE. Assuming the variables with higher orders of magnitude are not necessarily more important than those with lower orders of magnitude in your unsupervised dataset, scaling is typically necessary before clustering the data with a K-Means model.

3(g) FALSE. The DB-SCAN model requires the number of clusters as an input.

3(h) TRUE. If all cluster probabilities become 0 or 1 and all covariance matrices become diagonal, the EM algorithm (for a GMM) becomes equivalent to Lloyd's algorithm.

Chain: $f_1(X) \to \hat{y}_1$; $f_2(X, \hat{y}_1) \to \hat{y}_2$; …, for L targets. **Hamming loss** = mean of $(1 - \text{accuracy}_i)$, where accuracy_i is the fraction of labels correct within point i. Weighted metrics are not on the exam.

Q4 SHORT ANSWER - QUESTIONS + SOLUTIONS

4(a) [8 points] In the gradient vector for the Mean Squared Error (MSE) cost, the partial derivative with respect to w_j (j = 1, …, N) is:

$\partial\text{MSE}/\partial w_j = -(2/M)\sum_i(y_i - w^T x_i)x_{ij}$.

Explain where the factor −2 and the term x_{ij} come from.

Answer: Start from MSE = $(1/M)\sum_i(y_i - w^T x_i)^2$. The outer derivative of the square gives $2(y_i - w^T x_i)$. The chain rule supplies the inner derivative $-x_{ij}$, since $w_j x_{ij}$ is the only term of $w^T x_i$ that depends on w_j.

$\partial(y_i - w^T x_i)^2/\partial w_j = 2(y_i - w^T x_i)(-x_{ij})$.

4(b) [10 points] The derivative of the sigmoid function (with respect to its only variable) can be written as:

$\sigma'(x) = e^{-x}/(1 + e^{-x})^2$.

Explain/Formulate how the right side of the equation above can be split into two factors and express the factors in terms of σ(x).

Answer: Split the squared denominator:

$\sigma'(x) = [1/(1 + e^{-x})]\cdot[e^{-x}/(1 + e^{-x})]$.

The first factor is σ(x). In the second, add and subtract 1 in the numerator:

$e^{-x}/(1 + e^{-x}) = [(1 + e^{-x}) - 1]/(1 + e^{-x})$

$= 1 - 1/(1 + e^{-x}) = 1 - \sigma(x)$.

Therefore $\sigma'(x) = \sigma(x)[1 - \sigma(x)]$.

4(c) [9 points] Explain what kind of dataset a K-Prototypes clustering model is useful for. For a K-Prototypes clustering model, find the matching dissimilarity between the categorical parts of the following two observations: **[red, large, metal]** and **[red, small, wood]**. Lastly, explain why the total distance equation for K-Prototypes multiplies the matching dissimilarity by a factor γ.

Answer: K-Prototypes is useful for a **mix of numeric and categorical variables**. Matching dissimilarity counts categorical mismatches:

d = 0 (red = red) + 1 (large ≠ small)

+ 1 (metal ≠ wood) = **2**.

Total distance = Euclidean distance + γ·d.

The categorical distance is a nonnegative integer; the numeric Euclidean distance can be at any order of magnitude. **γ rescales one component so neither dominates** the total distance.

4(d) [8 points] The test confusion matrix for a three-class classification task is as follows:

Actual ↓ / Predicted →	Cat	Dog	Bird
Cat	20	3	2
Dog	4	15	1
Bird	6	2	12

Express the (One-vs-Rest/binary) precision and recall for the class **Bird** as fractions. You are not required to perform any calculations. (It is fine if you do.)

Answer: TP = 12 (Bird diagonal). FP = 2 + 1 = 3 (other entries in the Bird **column**). FN = 6 + 2 = 8 (other entries in the Bird **row**).

$\textbf{Precision}_{\textbf{Bird}}$ = TP/(TP + FP) = 12/(12 + 3)

= **12/15** (= 4/5).

$\textbf{Recall}_{\textbf{Bird}}$ = TP/(TP + FN) = 12/(12 + 8)

= **12/20** (= 3/5).

REVIEW: DBSCAN COURSE ALGORITHM

Start at a random point. Add the nearest point if its distance to the cluster is < tol; otherwise finalize and start a new cluster. Groups below min_samples become noise. Small tol changes can abruptly change the cluster count. Nonparametric: no Gaussian/source PDF is specified.